



Sébastien MARTINET

10/07/1996
French

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Languages

English	● ● ● ●
French (native)	● ● ● ●
Spanish	● ● ○ ○
Italian	● ● ○ ○

Numerical tools

Python	● ● ● ●
Fortran	● ● ● ●
C/C++	● ● ○ ○
R	● ● ○ ○
hTl5 & CSS	● ● ○ ○

Machine Learning & Data Processing

Pandas	● ● ● ●
SQL	● ● ● ○
scikit-learn	● ● ● ○
PyTorch	● ● ○ ○
TensorFlow	● ○ ○ ○
PySpark	● ● ○ ○

Education

- 2022 PhD in Astrophysics**
University of Geneva,
Switzerland
- 2018 Master in Astrophysics**
Université Grenoble-Alpes,
France
- 2016 Bachelor Fundamental
Physics**
Université Pierre et Marie
Curie, Paris VI, France

Experience

- Oct. 2022 – Present. **PostDoctoral Researcher in Nuclear
Astrophysics** [Université Libre de Bruxelles, BELGIUM](#)
- Sept. 2022 – Oct. 2022. **PostDoctoral Researcher in Stellar
Astrophysics** [University of Geneva, SWITZERLAND](#)
- Sept. 2018 – Aug. 2022. **Doctoral Researcher in Stellar
Astrophysics** [University of Geneva, SWITZERLAND](#)

Data Science Projects

Statistical Modeling & Scalable Data Pipelines [Python](#), [Fortran](#),
[HPC](#), [Git](#) (7 publications - 3 first-author)

- Built a scalable pipeline to run and analyze large-scale simulations (10k+ runs) using legacy Fortran code wrapped in Python; automated deployment on HPC clusters.
- Developed and applied a statistical framework to quantify and propagate uncertainties, enabling robust predictions from high-dimensional data.
- Delivered reproducible workflows and end-to-end tooling to extract actionable insights from complex simulation output.
- Collaborated with multidisciplinary teams to translate technical results into usable observables across domains.

Enhancing Legacy Software with Modern Data Science [Python](#),
[Fortran](#), [Data Viz](#), [CLI Tools](#) (6 publications - 2 first-author)

- Led the design and integration of new physics modules into a legacy simulation code, improving performance and expanding capabilities for broader use cases.
- Modernized the data analysis interface with Python tooling (NumPy, pandas, seaborn, Power BI), improving maintainability and enabling rapid visual exploration.
- Developed robust CLI tools and notebooks for data wrangling and visualization, enhancing reproducibility and team collaboration.
- Demonstrated autonomy and leadership in driving complex projects from concept to publication.

Other projects (55 collaborators, 19 institutes, 11 countries) include: integrating models with large observational datasets, analyzing experimental data to improve simulations, publishing data products, and building scientific data platforms.

Selected Publications

Martinet, S. et al. **(2025)** Statistical framework for nuclear parameter uncertainties in nucleosynthesis modeling of r- and i-process.
doi: [2025EPJA...61...48M](#)

Martinet, S. et al. **(2024)** The intermediate neutron capture process: IV. Impact of nuclear model and parameter uncertainties.
doi: [2024A&6A...684A...8M](#)

[Full list of publications](#) / [Full list of conferences](#)
(22 papers incl. 9 first author) / (15 talks in 10 different countries)